

The OSI Model & TCP/IP Stack

The 7-Layer Model

Layer	Name	Unit	Example Devices/Protocols
7	Application	Data	HTTP, DNS, SMTP
6	Presentation	Data	TLS handshake, JPEG, SSL
5	Session	Data	NetBIOS, RPC
4	Transport	Segment	TCP, UDP
3	Network	Packet	IP, ICMP, routing
2	Data Link	Frame	Ethernet, 802.11 (Wi-Fi)
1	Physical	Bits	Cables, radio signals

You Don't Need to Memorize the Layers – You Need to THINK in Layers

Most security problems are layer confusion:

- Phishing is layer 7 (it attacks the human)
- ARP spoofing is layer 2 (forged MAC mapping)
- IP spoofing is layer 3 (forged source IP)
- SYN flood is layer 4 (TCP handshake abuse)
- Route injection is layer 3 (BGP)

TCP/IP: The Real Stack

The OSI model is theory. The internet runs on TCP/IP – essentially 4 layers:

